

# Vector Calculus, Linear Algebra and Differential Forms: A Unified Approach - Solutions

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# Chapter 0

## Preliminaries

### 0.2 Quantifiers and Negation

**Exercise 0.2.1** Negate the following statements:

- Every prime number such that if you divide it by 4 you have a remainder of 1 is the sum of two squares.
- For all  $x \in \mathbb{R}$  and for all  $\epsilon > 0$ , there exists  $\delta > 0$  such that for all  $y \in \mathbb{R}$ , if  $|y - x| < \delta$ , then  $|y^2 - x^2| < \epsilon$ .
- For all  $\epsilon > 0$ , there exists  $\delta > 0$  such that for all  $x, y \in \mathbb{R}$ , if  $|y - x| < \delta$ , then  $|y^2 - x^2| < \epsilon$ .

**SOLUTION**

- There exists a prime number in the form of  $4n + 1$ , for  $n \in \mathbb{N}_0$  that cannot be expressed as the sum of two squares.
- There exists  $x \in \mathbb{R}$  and  $\epsilon > 0$  such that for all  $\delta > 0$ , there exists  $y \in \mathbb{R}$  with  $|y - x| < \delta$  but  $|y^2 - x^2| \geq \epsilon$ .
- There exists  $\epsilon > 0$  such that for all  $\delta > 0$ , there exists  $x, y \in \mathbb{R}$  with  $|y - x| < \delta$  but  $|y^2 - x^2| \geq \epsilon$ .

**Exercise 0.2.2** Explain why one of these statements is true and the other is false:

$$\begin{aligned} &(\forall \text{ man } M)(\exists \text{ woman } W) \mid W \text{ is the mother of } M \\ &(\exists \text{ woman } W)(\forall \text{ man } M) \mid W \text{ is the mother of } M \end{aligned}$$

**SOLUTION**

The difference between the two statements lies in the quantifier. Obviously the first statement is true while the second is false.

The first statement states that for every man, we can find a woman that is the man's mother. This is inherently true as every man must have a mother.

On the other hand, the second statement states that we can find a woman that is the mother of every man, which is obviously not possible.

### 0.3 Set Theory

**Exercise 0.3.1** Let  $E$  be a set, with subsets  $A \subset E$  and  $B \subset E$ , and let  $*$  be the operation  $A * B = (E - A) \cap (E - B)$ . Express the sets

$$\text{a. } A \cup B \quad \text{b. } A \cap B \quad \text{c. } E - A$$

using  $A$ ,  $B$  and  $*$ .

**SOLUTION**

Before solving this problem, consider the special case of the operator  $*$ :

$$X * X = (E - X) \cap (E - X) = X^c \cap X^c = X^c$$

- a. We can first simplify the operator with  $A * B = A^c \cap B^c = (A \cup B)^c$ . Using the special case, we get

$$(A * B) * (A * B) = (A \cup B)^c * (A \cup B)^c = A \cup B$$

- b. Working backwards, we get

$$A \cap B = (A^c \cup B^c)^c = ((A * A) \cup (B * B))^c = (A * A) * (B * B)$$

- c. This case is the equivalent to the special case above, hence  $E - A = A^c = A * A$ .

## 0.4 Functions

**Exercise 0.4.1** Are the following true functions? That is, are they both everywhere defined and well defined?

- "The aunt of", from people to people.
- $f(x) = \frac{1}{x}$ , from real numbers to real numbers.
- "The capital of", from countries to cities (careful - at least one country, Bolivia, has two capitals.)

**Exercise 0.4.2** a. Make up a nonmathematical function that is onto but not one to one.

- b. Make up a mathematical function that is onto but not one to one.

**Exercise 0.4.3** a. Make up a nonmathematical function that is bijective (onto and one to one).

- b. Make up a mathematical function that is bijective.

**Exercise 0.4.4** a. Make up a nonmathematical function that is one to one but not onto.

- b. Make up a mathematical function that is one to one but not onto.

**Exercise 0.4.5** Given the functions  $f : A \rightarrow B$ ,  $g : B \rightarrow C$ ,  $h : A \rightarrow C$ , and  $k : C \rightarrow A$ , which of the following compositions are well defined? For those that are, give the domain and codomain of each.

- |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|
| a. $f \circ g$ | b. $g \circ f$ | c. $h \circ f$ | d. $g \circ h$ | e. $k \circ h$ | f. $h \circ g$ |
| g. $h \circ k$ | h. $k \circ g$ | i. $k \circ f$ | j. $f \circ h$ | k. $f \circ k$ | l. $g \circ k$ |

**Exercise 0.4.6** Prove Proposition 0.4.11. **Inverse image of intersection, union**

- a. The inverse image of an intersection equals the intersection of the inverse images:

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$$

- b. The inverse image of a union equals the union of the inverse images:

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$$

**Exercise 0.4.7** Evaluate  $(f \circ g \circ h)(x)$  at  $x = a$  for the following.

- $f(x) = x^2 - 1$ ,  $g(x) = 3x$ ,  $h(x) = -x + 2$ , for  $a = 3$ .
- $f(x) = x^2$ ,  $g(x) = x - 3$ ,  $h(x) = x - 3$ , for  $a = 1$

**Exercise 0.4.8** a. Prove Proposition 0.4.16. **(Composition of onto functions)**. Let the functions  $f : B \rightarrow C$  and  $g : A \rightarrow B$  be onto. Then the composition  $(f \circ g)$  is onto.

- b. Prove Proposition 0.4.17. **(Composition of one to one functions)**. Let  $f : B \rightarrow C$  and  $g : A \rightarrow B$  be one to one. Then the composition  $(f \circ g)$  is one to one.

**Exercise 0.4.9** What is the natural domain of  $\sqrt{\frac{x}{y}}$ ?

**Exercise 0.4.10** What is the natural domain of

- a.  $\ln \circ \ln$ ?      b.  $\ln \circ \ln \circ \ln$ ?      c.  $\ln$  composed with itself  $n$  times?

**Exercise 0.4.11** What subset of  $\mathbb{R}$  is the natural domain of the function  $(1+x)^{1/x}$ ?

**Exercise 0.4.12** The function  $f(x) = x^2$  from real numbers to real nonnegative numbers is onto but not one to one.

- a. Can you make it one to one by changing its domain? By changing its codomain?  
b. Can you make it not onto by changing its domain? Its codomain?